

Each criterion was evaluated against every other using a pairwise comparison method. Values were assigned in 0.25 increments between 0 and 1, where:

- A value of 0.5 indicates equal importance.
- Values of 0.25 or 0.75 indicate one criterion is moderately more important than the other.
- Values of 0 or 1 reflect extreme preference toward one criterion, rendering the other negligible.

The comparisons are symmetric such that the sum of each pair equals 1. Total weights were summed and normalized to create priority values for each criterion.

Criteria	Mass	Power Demand	Complexity/	Radiation Durability	Reliability/TRL	Weight	Normalized
Mass		0.25	0.5	0.25	0.25	1.25	0.13
Power Demand	0.75		0.75	0.5	0.5	2.5	0.25
Complexity/ Integration	0.5	0.25		0.25	0.25	1.25	0.13
Radiation Durability	0.75	0.5	0.75		0.5	2.5	0.25
Reliability/TRL	0.75	0.5	0.75	0.5		2.5	0.25
Total						10	

	Mass	Power Demand	Complexity/	Radiation Durability	Reliability/TRL
5	< 5 kg added	< 25 W	Simple installation,	Flight-proven; negligible	TRL 8-9
4	5-9 kg	25-50 W	Moderate integration	Minor degradation;	TRL 6-7
3	10-14 kg	50-100 W	Custom mounts	Some degradation;	TRL 5-6
2	15-20 kg	100-150 W	Multiple interfaces /	Noticeable degradation /	TRL 4-5
1	> 20 kg	> 150 W	New mechanisms /	Strongly affected /	TRL < 3

Analysis of Night Survival Thermal Architecture							
Decision Criteria		MLI and heaters		Aerogel and heaters		PCM with heat pipes	
Criteria	Weight	Score	Weighted Score	Score	Weighted Score	Score	Weighted Score
Mass	0.13	4	0.52	4	0.52	2	0.26
Power Demand	0.25	5	1.25	3	0.75	4	1.00
Complexity/ Integration	0.13	4	0.52	3	0.39	2	0.26
Radiation Durability	0.25	5	1.25	3	0.75	3	0.75
Reliability/TRL	0.25	5	1.25	3	0.75	2	0.50
Total Score		4.79		3.16		2.77	